

Introduction to Amazon EC2

Launch • Monitor • Security Groups • Resize • Stop & Termination Protection

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✓ **LAB RESULT:** 25 / 25 — full marks on all graded checks (instance, system log, security group, resize, stop protection).

Lab Overview & Objectives

AWS Academy — Introduction to Amazon EC2 • Duration ~35 min

Amazon EC2 provides resizable compute capacity in the cloud. This lab launches a web server from a bootstrap script, monitors it, opens it to HTTP traffic through a security group, resizes both the instance and its disk, explores service limits, and exercises the guardrails that prevent accidental stop and termination.

Objectives

- Launch a web server with termination protection enabled.
- Monitor the EC2 instance (status checks, CloudWatch, system log, screenshot).
- Modify the security group to allow HTTP (port 80) access.
- Resize the instance type and EBS volume, and enable stop protection.
- Explore EC2 service limits, then test stop protection and stop the instance.

Architecture

A single EC2 web server behind a security group

One EC2 instance runs in a public subnet of the Lab VPC, inside a security group that acts as its virtual firewall. A user-data bootstrap script turns it into a working web server on first boot.



Figure 1 — EC2 web server inside a security group in one Availability Zone.

CORRECTION

The AWS diagram labels the server “IIS Web Server,” but that’s inaccurate for this build. IIS is Microsoft’s Windows web server; this instance runs **Amazon Linux 2023** and the user-data script installs **Apache (httpd)**. Knowing your actual stack matters — the running service is Apache, not IIS.

Task 1 — Launch Your EC2 Instance

With termination protection and a user-data bootstrap

1. EC2 (region [us-east-1](#) / [N. Virginia](#)) → **Launch instance**.
2. Configure:

Launch configuration

Name:	Web Server	(creates tag Name = Web Server)
AMI:	Amazon Linux 2023	
Instance type:	t2.micro	(1 vCPU, 1 GiB)
Key pair:	vockey	
Network:	Lab VPC / PublicSubnet1	(auto-assign public IP)
Security group:	Web Server security group	(remove the default inbound rule)
Storage:	8 GiB root (default)	
Advanced:	Termination protection = Enable	

3. Paste the bootstrap script into **User data**:

User data – runs as root on first boot

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1></html>' > /var/www/html/index.html
```

4. Launch, then wait for [Running](#) and [2/2 checks passed](#).

SECURITY INSIGHT

User data is instance bootstrapping: the script runs once, as root, on first boot — installing Apache, enabling it at startup, and publishing a page with no manual SSH. **Termination protection** is a safety control that blocks accidental deletion of the instance while enabled.

SCREENSHOT

Web Server instance in the Running state with 2/2 status checks passed.

Task 2 — Monitor Your Instance

Status checks, CloudWatch, system log, screenshot

1. **Status checks** tab — confirm **System reachability** and **Instance reachability** both pass.
2. **Monitoring** tab — CloudWatch metrics (basic 5-minute monitoring is on by default; detailed 1-minute is optional).
3. Actions → Monitor and troubleshoot → **Get system log** — verify the [httpd](#) package installed from the user-data script.
4. Actions → Monitor and troubleshoot → **Get instance screenshot** — a console screenshot for troubleshooting when SSH/RDP is unavailable.

NOTE

The system log and instance screenshot are diagnosis tools for when an instance is unreachable before its SSH daemon starts — they give visibility into boot and kernel problems without needing to connect in.

 SCREENSHOT

System log output confirming the httpd package was installed via user data.

 SCREENSHOT

Instance screenshot from Get instance screenshot.

Task 3 — Security Group & Web Access

Open port 80 to reach the site

1. Copy the instance's **Public IPv4 address** and open it in a browser — it **fails to load**.
2. Security Groups → **Web Server security group** → **Inbound rules** → Edit → Add rule:

Inbound rule

Type: HTTP
Port: 80
Source: Anywhere-IPv4 (0.0.0.0/0)

3. Save, refresh the browser — the page now shows **Hello From Your Web Server!**.

SECURITY INSIGHT

This is the security-group-as-firewall lesson in miniature: the web server was running the whole time, but unreachable because **no inbound rule permitted port 80**. Deny-by-default means nothing gets in until you explicitly allow it — the failed load, then the fix, makes the control visible.

SCREENSHOT

Browser showing "Hello From Your Web Server!" after adding the HTTP inbound rule.

Task 4 — Resize the Instance & Volume

Scale compute and storage; enable stop protection

1. Stop the instance (required before resizing). No runtime charge while stopped; EBS storage still bills.
2. Actions → Instance settings → **Change instance type**: **t2.micro** → **t2.small** (2× the memory).
3. Actions → Instance settings → **Change stop protection** → Enable → Save.
4. Storage tab → select the volume → Actions → **Modify volume**: **8 GiB** → **10 GiB**, confirm.
5. Start the instance again — now t2.small with a 10 GiB root volume.

NOTE

On stop/start, an instance is usually migrated to a new host and gets a **new public IPv4**, but keeps its private IP and all EBS data. Elasticity works in both directions — you can size compute and storage up or down as the workload changes.

SCREENSHOT

Instance details after resize — t2.small with a 10 GiB root volume.

Task 5 — Explore EC2 Limits

Service Quotas (not graded)

1. Service Quotas → AWS services → **Amazon Elastic Compute Cloud (Amazon EC2)**.
2. Search **running on-demand** and observe the per-region limits on how many/which instance types can run.

NOTE

Default service quotas exist per region for images, instances, volumes, and snapshots. Launches that would exceed a running-instance limit are rejected — account owners can request increases. (This task carries no graded check.)

Task 6 — Test Stop Protection

The guardrail blocks the first stop

1. With stop protection enabled, choose **Stop instance** — it **fails**: *The instance may not be stopped. Modify its 'disableApiStop' attribute and try again..*
2. Actions → Instance settings → **Change stop protection** → uncheck Enable → Save.
3. Stop the instance again — it now **succeeds**.

SECURITY INSIGHT

Stop protection (and termination protection) are deliberate friction against destructive mistakes: the `disableApiStop` attribute forces an explicit, conscious step before a stop or delete can go through — a small control that prevents costly accidents.

GRADER TIMING

Some checks in this lab only credit **~5 minutes after** the action is complete. Submitting too early can score a task 0/5 even when done correctly — wait a couple of minutes and re-submit. (This is exactly what caught Task 6 on the first attempt; it scored 5/5 on re-submission.)



SCREENSHOT

Stop failing on the first attempt (stop protection), then succeeding after it's disabled.

Security Insights & Key Takeaways

Notes for the CYB 222 capstone article

- **Security groups are deny-by-default firewalls.** The site was unreachable until an explicit HTTP:80 inbound rule was added — the clearest demo of allow-listing network access.
- **User data automates secure bootstrapping.** The instance configured itself on boot (install, enable, start, publish) with no interactive login — repeatable and hands-off.
- **Guardrails prevent costly mistakes.** Termination and stop protection add deliberate friction against accidental destruction of running resources.
- **Know your actual stack.** The diagram said IIS; the box ran Apache on Amazon Linux. Accurate mental models matter when you're securing or troubleshooting a system.
- **Elasticity, up and down.** Instance type and EBS volume both resized on demand; stop/start rotates the public IP but preserves data and the private IP.
- **Know the limits.** Per-region service quotas cap running instances — relevant for both capacity planning and spotting abnormal (possibly malicious) resource growth.

Lab Result

AWS Academy submission report

All graded checks passed — instance creation, system log, security group, resize, and stop protection. (Task 5, Explore EC2 Limits, has no graded check.)

Total score	25/25
Task 1 - EC2 instance created correctly	5/5
Task 2 - get system log requested	5/5
Task 3 - security group updated	5/5
Task 4 - EC2 instance updated	5/5
Task 6 - Instance stopped on second try	5/5

Submission report — Total score 25 / 25.